

Initial Design Document

Track: Design Provocation from Syllabus

Community Partner: N/A

Project Title: Accessible AI Tool Explainer for Makerspaces

Group Members: Saeed AlKaabi, Azan Malik, Matus Menczer, Joseph Zhao

Design Problem (around 200-300 words)

Many makerspaces rely heavily on visual instructions, technical terminology, and prior experience when teaching users how to use tools and equipment. This creates barriers for beginners and can especially exclude users with visual impairments or limited technical experience. Even simple tools such as laser cutters, soldering stations, or 3D printers often require users to interpret visual labels, safety diagrams, or complicated manuals before using them safely. As discussed in class readings related to accessibility and mismatch, many environments unintentionally assume a “default user,” which can make participation more difficult for others.

Our project explores how artificial intelligence and multimodal interaction can make makerspaces more accessible and beginner-friendly. We aim to create a prototype where users can upload or capture an image of a makerspace tool, and the system will identify the object and provide simplified explanations, safety instructions, and usage guidance through text. The project focuses on reducing dependence on others, providing users with a safe space and access to tools, and offering a text-to-speech solution for makers with visual impairments, learning differences, or limited prior technical experience.

This is a design challenge because accessibility needs differ across users, and AI-generated explanations may not always be accurate or understandable. Our team will need to balance simplicity, accessibility, safety, and usability while creating a system that feels intuitive and supportive for new makerspace participants.

Plan for Addressing Design Problem

Timeline: What activities and deliverables are you planning for the next three weeks? Aim to set reasonable goals for your team.

Week 1: (What have you accomplished?)

- Brainstormed multiple project ideas related to accessibility and makerspaces

- Selected the AI Tool Explainer concept
- Discussed core features and accessibility goals
- Began researching multimodal interaction and accessible design approaches
- Planned initial frontend structure and API integration

Week 2:

- Develop a working React frontend
- Implement image upload functionality
- Connect Gemini API for tool identification
- Generate AI explanations and safety instructions
- Add basic text-to-speech support
- Begin early testing with classmates

Week 3:

- Refine interface and accessibility features
- Improve prompts and AI response formatting
- Conduct informal user testing
- Record demonstration video
- Complete final report and presentation materials

Potential Solutions: What are potential solutions your team has considered to address the design problem? Please describe 3 solutions or approaches your team has discussed and how they address the problem below. Feel free to include sketches if appropriate.

Solution 1: AI Image-Based Tool Explainer

This solution allows users to upload an image of a makerspace tool and receive an AI-generated explanation about what the tool is, how it is used, and important safety precautions. The system can also provide beginner-friendly step-by-step guidance and optional audio playback to support multimodal interaction. This approach is accessible for beginners, scalable, and reduces dependence on visual manuals or prior technical knowledge. However, the system may occasionally misidentify tools depending on image quality and requires internet/API access to function properly.

Solution 2: Voice-Based Makerspace Assistant

This approach focuses entirely on voice interaction. Users can ask spoken questions about makerspace tools, safety procedures, or instructions, and the system responds using speech output. This creates a more hands-free and accessibility-focused experience, especially for visually impaired users or users who may struggle with reading technical documentation. While this solution strongly supports accessibility goals, it introduces challenges such as speech recognition inaccuracies and more complex technical implementation. It may also be harder to visually demonstrate during presentations and testing.

Solution 3: Interactive Audio Navigation System

This concept explores accessibility through spatial navigation within a makerspace. Users would interact with a digital map that provides spoken descriptions, audio cues, and location-based guidance to help them move through unfamiliar environments and identify tools or workstations. The system encourages nonvisual interaction and supports users who may struggle with traditional visual navigation systems. However, accurately mapping physical makerspaces and maintaining reliable navigation would require additional technical complexity and would be more difficult to prototype within the limited course timeline.

Mid-project Update

Track:

Community Partner

Project Title

Group Members

Project Progress (around 400-500 words)

What progress has your group made since starting the project? What are potential challenges you foresee for solving your design problem? Describe in detail how you will iterate on your design in the next two weeks to help you overcome these challenges. You should also describe any activities you will do to help you iterate on your design (e.g., user testing, interviews, literature review).

Timeline

What activities and deliverables are you planning for the next two weeks? If you made changes to your timeline, please describe it below, too.

Week 2: (What have you accomplished?)

Week 3:

Final Submission(s)

Project Title

Group Members

Target Conference (optional):

You are almost at the end of this assignment! You will now write a final report in the form of a *written submission* and *video presentation* of 3-5 minutes.

Your group might consider submitting this work to a conference. The teaching team can offer some recommendations on this if this is something you are interested in. Example conferences include: American Educational Research Association ([AERA](#)), Interaction Design and Children ([IDC](#)), International Conference of the Learning Sciences ([ICLS](#)), ACM SIGACCESS Conference on Computers and Accessibility (ASSETS) and Tangible Embodied Interaction ([TEI](#)).

Your final report will be an *extended abstract* (use [this template](#)). It should include roughly the following sections in no more than 4 pages. This format and page restrictions are intended to guide you in the writing process by including a standard set of headers and an explicit page limit. For students interested in academic publishing, using an this format will hopefully simplify the conference submission process. Please reach out to the teaching team if you have any questions: Here are the sections of the paper.

- Abstract (no more than 150 words): A short descriptive summary of your solution and how it addresses the design problem and contributes to prior research.
- Keywords (4-5 words/phrases): Words or short phrases to help readers find your project.
- Introduction and Related Work: Describe the motivation for the project, and connect it to a need or problem space by referring to prior research and any relevant theories. Briefly introduce your design solution and discuss how it solves this design need and contributes to prior research.
- Positionality Statement: Describe your positionality relative to this work and the community.
- Overview of Project “Deliverable”: Describe in more detail what you created for this final project. What are the essential components of the design?
- Design Process and Preliminary Evaluation: Describe your design process and iterations to theoretically and practically justify your solution. If you evaluated your design, be sure to describe what methods you used to test it, who you tested it with, and what you learned.
- Conclusions and Future Work: What can you conclude about your design and how does it contribute to prior research? How does it contribute to and/or reflect the community that you worked with? What additional work needs to be done in the future?
- References: Use the ACM Style to cite your sources (see template)